CCNA 200-301 v1.1

Official Syllabus · Categorised Topics · 10-Day Study Plan

Based on the Official Cisco Exam Topics (2024 release)

Exam Code	200-301 v1.1	Duration	120 minutes
Questions	~50–60	Passing Score	~800 / 1000
Validity	3 years	Provider	Pearson VUE

■ PART 1 — CATEGORISED SYLLABUS (6 Domains)

■ 1.0 Network Fundamentals

Exam Weight: 20%

1.1	**Role & function of network components**	Routers · Layer 2/3 Switches · NG Firewalls & IPS · Access Points · Controllers · Endpoints · Servers · PoE
1.2	**Network topology architectures**	Two-tier · Three-tier · Spine-leaf · WAN · SOHO · On-premises vs Cloud
1.3	**Physical interfaces & cabling**	Single-mode fibre · Multimode fibre · Copper · Ethernet (shared media & point-to-point)
1.4	**Interface & cable issues**	Collisions · Errors · Duplex mismatch · Speed mismatch
1.5	**TCP vs UDP**	Comparison of transport-layer protocols
1.6–1.9	**IPv4 & IPv6 Addressing**	IPv4 subnetting · Private addressing · IPv6 config · IPv6 types (Unicast/Anycast/Multicast/EUI-64)
1.10	**Verify IP – Client OS**	Windows · macOS · Linux
1.11	**Wireless principles**	Non-overlapping channels · SSID · RF · Encryption
1.12	**Virtualisation fundamentals**	Server virtualisation · Containers · VRFs
1.13	**Switching concepts**	MAC learning/aging · Frame switching · Frame flooding · MAC address table

■ 2.0 Network Access

Exam Weight: 20%

2.1	**VLANs**	Access ports (data & voice) · Default VLAN · Inter-VLAN connectivity
2.2	**Interswitch connectivity**	Trunk ports · 802.1Q · Native VLAN
2.3	**Layer 2 discovery protocols**	Cisco Discovery Protocol (CDP) · LLDP
2.4	**EtherChannel (LACP)**	Layer 2 & Layer 3 EtherChannel config & verification

2.5	**Rapid PVST+ STP**	Root port/bridge · Port states & roles · PortFast · Root guard · Loop guard · BPDU filter/guard
2.6	**Cisco Wireless Architectures**	AP modes: Local, FlexConnect, Monitor, Sniffer, etc.
2.7	**WLAN physical infrastructure**	AP · WLC · Access/trunk ports · LAG
2.8	**Device management access**	Telnet · SSH · HTTP/HTTPS · Console · TACACS+ / RADIUS · Cloud managed
2.9	**Wireless LAN GUI config**	WLAN creation · Security settings · QoS profiles · Advanced settings

■■ 3.0 IP Connectivity

Exam Weight: 25%

3.1	**Routing table components**	Protocol code · Prefix · Network mask · Next hop · Admin distance · Metric · Gateway of last resort
3.2	**Router forwarding decision**	Longest prefix match · Administrative distance · Routing protocol metric
3.3	**Static routing (IPv4 & IPv6)**	Default route · Network route · Host route · Floating static route
3.4	**OSPFv2 (single area)**	Neighbour adjacencies · Point-to-point · Broadcast (DR/BDR) · Router ID
3.5	**First Hop Redundancy Protocols**	Purpose, functions & concepts of FHRP (HSRP, VRRP, GLBP)

■■ 4.0 IP Services

Exam Weight: 10%

4.1	**NAT**	Inside source NAT — static & pool configuration
4.2	**NTP**	Client and server mode configuration
4.3	**DHCP & DNS**	Role within the network
4.4	**SNMP**	Function in network operations
4.5	**Syslog**	Facilities and severity levels
4.6	**DHCP client & relay**	Configuration and verification
4.7	**QoS (PHB)**	Classification · Marking · Queuing · Congestion · Policing · Shaping
4.8	**SSH remote access**	Configure devices for SSH
4.9	**TFTP / FTP**	Capabilities and functions in the network

■ 5.0 Security Fundamentals

Exam Weight: 15%

5.1	**Key security concepts**	Threats · Vulnerabilities · Exploits · Mitigation techniques
5.2	**Security program elements**	User awareness · Training · Physical access control
5.3	**Device access control**	Local passwords configuration & verification
5.4	**Password policy elements**	Management · Complexity · MFA · Certificates · Biometrics
5.5	**VPNs**	IPsec remote access and site-to-site VPNs
5.6	**Access Control Lists (ACLs)**	Configuration and verification

5.7	**Layer 2 security**	DHCP snooping · Dynamic ARP inspection · Port security
5.8	**AAA concepts**	Authentication · Authorisation · Accounting
5.9	**Wireless security protocols**	WPA · WPA2 · WPA3
5.10	**WLAN with WPA2 PSK**	GUI configuration and verification

■ 6.0 Automation & Programmability

Exam Weight: 10%

6.1	**Automation impact**	How automation impacts network management
6.2	**Traditional vs controller-based**	Comparison of networking models
6.3	**SDN architecture**	Overlay · Underlay · Fabric · Control/data plane separation · Northbound & Southbound APIs
6.4	**AI & ML in networking**	Generative AI · Predictive AI · Machine learning in network operations
6.5	**REST-based APIs**	Authentication types · CRUD · HTTP verbs · Data encoding
6.6	**Config management tools**	Ansible · Terraform
6.7	**JSON-encoded data**	Recognise components and structure

■ PART 2 — 10-DAY CCNA STUDY PLAN

Each day = approx. 4–5 hours of focused study. Adjust pace as needed. Practical lab time is highlighted separately.

DAY 1 | Network Fundamentals — Concepts & Components

Domain 1.0 (part 1) | Weight: 20%

■ Topics to Cover

- Role of routers, switches (L2/L3), firewalls, IPS, APs, controllers, servers
- Topology architectures: two-tier, three-tier, spine-leaf, WAN, SOHO, cloud
- Physical media: fibre (single/multi-mode), copper, Ethernet connections
- Interface issues: collisions, errors, duplex & speed mismatch
- TCP vs UDP — when each protocol is used

■■ Lab Exercise	■ Study Tip
Packet Tracer: Build a 3-tier topology; observe MAC table with 'show mac address-table'	*Use the OSI model as your mental map — every topic anchors to a specific layer.*

■ *Resources: Cisco Press CCNA OCG Ch.1–3 | Packet Tracer labs*

DAY 2 | IPv4, IPv6 Addressing & Subnetting

Domain 1.0 (part 2) | Weight: 20%

■ Topics to Cover

- IPv4 addressing, CIDR, subnetting (calculating hosts, networks, broadcast)
- Private IPv4 ranges (RFC 1918)
- IPv6 config & prefix notation
- IPv6 types: Unicast (global, unique local, link-local), Anycast, Multicast, EUI-64
- Verify IP parameters on Windows / macOS / Linux

■■ Lab Exercise	■ Study Tip
Subnetting practice: divide 192.168.10.0/24 into 8 subnets; configure dual-stack on routers	*Master subnetting with the 'magic number' method — it's essential for every CCNA question block.*

■ *Resources: subnettingpractice.com | CCNA OCG Ch.12–14*

DAY 3 | Wireless & Switching Concepts + Virtualisation

Domain 1.0 (part 3) | Weight: 20%

■ Topics to Cover

- Wireless: non-overlapping channels (1/6/11), SSID, RF bands, encryption basics
- Switching concepts: MAC learning/aging, frame switching, flooding, MAC table

- Virtualisation: server virtualisation, containers (Docker), VRFs

Lab Exercise	Study Tip
Configure a basic wireless network in Packet Tracer; explore MAC table aging	*Wi-Fi channels 1, 6, 11 are the three non-overlapping channels in 2.4 GHz — memorise them.*

Resources: CCNA OCG Ch.27–28 | Neil Anderson (Udemy) wireless section

DAY 4 | Network Access — VLANs, Trunks & STP

Domain 2.0 (part 1) | Weight: 20%

Topics to Cover

- VLANs: access ports (data & voice), default VLAN 1, inter-VLAN routing
- Interswitch connectivity: trunk ports, 802.1Q encapsulation, native VLAN
- CDP & LLDP — Layer 2 discovery protocol config
- EtherChannel with LACP — L2 and L3 modes
- Rapid PVST+: root bridge election, port states/roles, PortFast, BPDU guard/filter, root guard, loop guard

Lab Exercise	Study Tip
Configure VLANs, trunks, and STP; verify with 'show spanning-tree' and 'show vlan brief'	*Remember STP port roles: Root Port → best path to root; Designated Port → best on segment.*

Resources: CCNA OCG Ch.8–10 | Boson NetSim labs

DAY 5 | Network Access — Wireless LAN & Device Management

Domain 2.0 (part 2) | Weight: 20%

Topics to Cover

- Cisco wireless architectures & AP modes (Local, FlexConnect, Monitor)
- WLAN infrastructure: AP, WLC, access/trunk ports, LAG
- Device management: Telnet, SSH, HTTP/HTTPS, console, TACACS+/RADIUS, cloud-managed
- Wireless LAN GUI: WLAN creation, security, QoS profiles, advanced settings

Lab Exercise	Study Tip
Configure SSH on a Cisco IOS device; explore WLC GUI in Packet Tracer	*Always prefer SSH over Telnet in exam scenarios — it's encrypted.*

Resources: CCNA OCG Ch.29–30 | Cisco dCloud wireless lab

DAY 6 | IP Connectivity — Routing & OSPF

Domain 3.0 | Weight: 25%

Topics to Cover

- Routing table: protocol codes, prefix, mask, next-hop, AD, metric, gateway of last resort
- Router forwarding: longest prefix match, administrative distance, metric

■	Static routing (IPv4 & IPv6): default, network, host, floating static routes
■	OSPFv2 single area: neighbour adjacencies, point-to-point, broadcast (DR/BDR), Router ID
■	First Hop Redundancy Protocols (HSRP, VRRP, GLBP)

■■ Lab Exercise	■ Study Tip
Configure OSPF multi-router topology; verify with 'show ip ospf neighbor' and 'show ip route'	*AD values to memorise: Connected=0, Static=1, OSPF=110, RIP=120, EBGP=20.*

■ *Resources: CCNA OCG Ch.18–22 | Packet Tracer OSPF lab*

DAY 7 | IP Services — NAT, DHCP, DNS, NTP, QoS & More

Domain 4.0 | Weight: 10%

■ Topics to Cover

■	NAT: inside source static and pool configuration
■	NTP: client/server mode
■	DHCP role & DNS role in the network; DHCP client & relay config
■	SNMP function; Syslog — facilities & severity levels (0–7)
■	QoS PHB: classification, marking, queuing, congestion, policing, shaping
■	SSH for remote access; TFTP/FTP capabilities

■■ Lab Exercise	■ Study Tip
Configure NAT overload (PAT); set up a DHCP server and relay agent	*Syslog severity: 0=Emergency, 1=Alert, 2=Critical, 3=Error, 4=Warning, 5=Notice, 6=Info, 7=Debug.*

■ *Resources: CCNA OCG Ch.24–26 | Packet Tracer IP services lab*

DAY 8 | Security Fundamentals

Domain 5.0 | Weight: 15%

■ Topics to Cover

■	Key concepts: threats, vulnerabilities, exploits, mitigation techniques
■	Security program: user awareness, training, physical access control
■	Device access control: local passwords, password policies, MFA, certificates
■	VPNs: IPsec remote access and site-to-site
■	ACLs: standard and extended, config & verification
■	Layer 2 security: DHCP snooping, Dynamic ARP inspection, port security
■	AAA: Authentication, Authorisation, Accounting
■	Wireless security: WPA / WPA2 / WPA3; configure WLAN with WPA2 PSK via GUI

■■ Lab Exercise	■ Study Tip
Configure extended ACLs to permit/deny specific traffic; enable port security on switch	*ACL processing: top-down, first match wins — always put more specific entries first.*

■ *Resources: CCNA OCG Ch.31–35 | Cisco Packet Tracer security labs*

DAY 9 | Automation, Programmability & AI in Networking

Domain 6.0 | Weight: 10%

■ Topics to Cover

- Automation impact on network management
- Traditional vs controller-based networking
- SDN architecture: overlay, underlay, fabric; control/data plane separation; northbound & southbound APIs
- AI & ML in networking: generative AI, predictive AI, machine learning operations
- REST APIs: authentication types, CRUD operations, HTTP verbs, data encoding
- Configuration management: Ansible, Terraform
- JSON-encoded data — recognise components and structure

■■ Lab Exercise	■ Study Tip
Parse a sample JSON API response; review Ansible playbook structure (no CLI needed)	*SDN northbound API faces applications; southbound API (OpenFlow, NETCONF) faces devices.*

■ *Resources: CCNA OCG Ch.36–38 | YouTube: David Bombal SDN/Automation*

DAY 10 | Full Revision, Practice Exam & Weak-Area Drill

All Domains — Consolidation Day

■ Topics to Cover

- Morning (2 hrs): Full timed practice exam (Boson / Cisco official)
- Review all incorrect answers — map each to its domain
- Subnetting speed drill — aim for <10 sec per subnet question
- Quick-fire: OSPF, STP, VLANs, ACL, NAT config commands
- Afternoon: Revisit your weakest domain with targeted labs
- Evening: Flash-card review — AD values, port numbers, syslog levels, QoS markings

■■ Lab Exercise	■ Study Tip
Full mixed-topic Packet Tracer scenario covering VLANs, OSPF, NAT, ACL, and SSH	*Trust your preparation. Flag uncertain questions and revisit — don't change answers impulsively.*

■ *Resources: Boson ExSim | Cisco official practice exam | Anki flashcard deck*

■ QUICK REFERENCE — DOMAIN WEIGHTS & DAY MAPPING

Domain	Title	Weight	Study Day(s)
1.0	Network Fundamentals	20%	Day 1, 2, 3
2.0	Network Access	20%	Day 4, 5
3.0	IP Connectivity	25%	Day 6
4.0	IP Services	10%	Day 7
5.0	Security Fundamentals	15%	Day 8
6.0	Automation & Programmability	10%	Day 9
—	Full Revision & Mock Exam	—	Day 10

Recommended resources: Cisco Press CCNA Official Cert Guide (OCG) · Neil Anderson / Jeremy's IT Lab (YouTube/Udemy) · Boson ExSim practice exams · Cisco Packet Tracer / GNS3 · Cisco Learning Network community